Calculus of Variations: Euler–Lagrange Equations and Free End Points

Optimal Control, Homework 2

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The problems come from Chapter 4 of Kirk, Optimal Control Theory.

Problem 4-4 (first variation). For $J(x) = \int_{t_0}^{t_f} [x^2 - x^2\dot{x}]\, dt$ I expanded $J(x+\delta x) - J(x)$, dropped the terms of higher order in δx, and obtained the first variation $\delta J = \int_{t_0}^{t_f} (3x^2\delta x - 2x\dot{x}\,\delta x - x^2\delta\dot{x})\, dt$.

Problem 4-8 (fixed end points). I solved the Euler–Lagrange equation $\partial g/\partial x - \frac{d}{dt}\partial g/\partial \dot{x} = 0$ for several functionals with both end points fixed:

- $\int_0^1 (x^2 + \dot{x}^2)\, dt$ with $x(0) = 0$, $x(1) = 1$ gives $x^\star(t) = (e^t - e^{-t})/(e - e^{-1})$;
- a variant with $x(0) = 1$, $x(2) = -3$;
- a two-state problem whose fourth-order characteristic equation $s^4 - 1 = 0$ gives exponential and trigonometric modes.

Problem 4-9. A functional with a forcing term. The particular solution $x_p = A$ plus the homogeneous modes $e^{\pm 2t}$, fitted to $x(0) = 1$ and $x(1) = 4$.

Problem 4-10 (free final state). The natural boundary condition $\partial g/\partial \dot{x}\,|_{t_f} = 0$ replaces the missing end condition. Example: $\int_0^1 (\frac{1}{2}\dot{x}^2 + x\dot{x} + \dot{x} + x)\, dt$ with $x(0) = \frac{1}{2}$ gives $x^\star(t) = \frac{1}{2}t^2 - \frac{3}{2}t + \frac{1}{2}$.

Problems 4-13 and 4-14 (end point on a curve). For arc-length-type functionals $\int \sqrt{1 + \dot{x}^2}\, dt$ and $\int \sqrt{1 + \dot{x}^2}/x\, dt$ with the final point constrained to a line $\theta(t)$, I used the transversality condition $[\partial g/\partial \dot{x}\,(\dot{\theta} - \dot{x}) + g]_{t_f} = 0$. The answers are the straight line $x^\star(t) = \frac{1}{4}t + 2$ and the circular arc $x^\star(t) = \sqrt{10 - t^2}$.